Property	Type	Mandatory	Description
eventReasonCode	*String*	*no*	*Pre-defined event reason code*
eventReasonText	*String*	*no*	*Event reason description*
<Other custom parameter>		*no*	*Based on model definition*

Post Events

Request

Request URI

- For HTTP basic authentication

  https://<server address>/api/inbound/rest/v1/{model namespace}.{write service name}/{event name}

  #### Example

  https://*<server address>*/api/inbound/rest/v1/com.lbngttapps.gtt.app.gttft1.gttft1WriteService/Picking

HTTP Method

Post

Request Header

Pass additional information with the request

Request Header Properties:

Property	Type	Mandatory	Description
LBN-GTT-Source-Id	String	No	An id to represent which source system the message comes from
LBN-GTT-Message-Number	String	No	The message's id in the source system
LBN-GTT-Partition-Key-Hint	String	No	A hint to determine which message queue the message will be sent to. Messages with the same hint will be sent to the same message queue and they will be processed in chronological order.

Sample Code

Request header example:

```
LBN-GTT-Message-Number: 9900000007778950
LBN-GTT-Source-Id: QS8CLNT910
LBN-GTT-Partition-Key-Hint: 20210122
```

Request Body

Here are sample request body of different scenarios:

- Post an event without attachments
- Post an event with attachments
- Post an event with estimatedTimestamps

Response

Response Codes

The following HTTP response codes may occur:

Code	HTTP Description	Track and Trace Semantic
202	Accepted	Request has been received and will be processed later
400	Bad Request	The content of the request payload does not match the definition. The request payload should be qualified json string and contain altKey.
401	Unauthorized	There is no authentication passed with the request
403	Forbidden	The user does not access the right url to create the tracked process instance
404	Not found	The access url is inaccessible
405	Method not allowed	The method received in the request-line is known by the origin server but not supported by the target resource.
500	Internal Server Error	An unspecified error occurred during request processing
503	Service Unavailable	The request cannot be executed due to unavailability of required components, e.g. tenant onboarding hasn't been completed
504	Timeout	The server, while acting as a gateway or proxy, did not receive a timely response from an upstream server it needed to access in order to complete the request.

Response Body

The response body is empty.

Post an event without attachments

In this case, the request body is in json format.

Sample Code

```
{
    "altKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY_IT:0080045552000040
"quantity": 0.000,
"messageSourceType": "TM",
```

```
    "actualTechnicalTimestamp": "2020-08-03T05:50:30.183Z",
    "actualBusinessTimeZone": "CET",
    "actualBusinessTimestamp": "2020-08-03T05:50:11Z",
    "locationAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:Location:ShippingPoint:0001"
}
```

Post an event with attachments

You can post an event with attachments in two formats: form-data format and JSON format.

Form-data Format

Request Header

```
Content-Type: multipart/form-data
```

Request Body

Here is the sample request body:

Sample Code

```
event:
{
    "altKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY_IT:0080045552000040
    "quantity": 0.000,
    "messageSourceType": "TM",
    "actualTechnicalTimestamp": "2020-08-03T05:50:30.183Z",
    "actualBusinessTimeZone": "CET",
    "actualBusinessTimestamp": "2020-08-03T05:50:11Z",
    "locationAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:Location:ShippingPoint:0001",
    "myattachments": [
        {
            "fileName": "POD.png"
        },
        {
            "fileName": "PrivatePic.png",
            "privateAttachmentURL": "<URL to your private file>"
        }
    ]
}
attachment[]:file list
```

Note

When sending “POST” requests, set the request body to “form-data” and add two key-value pairs:

Key	Type	Value
event	Text	Copy the payload of “event”
attachments[]	File	Select the files with the same file name you defined in “myattachments”. They can't be private URLs.

- In this example, “`myattachments`” is an event field with the type “Attachments” you defined in your model. You can have user-defined names.
- You must **upload** at least one file. You can also add private URLs to attachments with `privateAttachmentURL`. However, sending a request with private attachment URLs only is not supported. You can use the JSON Format for such case.

JSON Format

Request Header

Set the `LBN-GTT-Input-Data-Attachments-Flag` to true.

Request Body

Here is the sample request body:

Sample Code

```
{
    "altKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY_IT:0080045552000040
    "quantity": 0.000,
    "messageSourceType": "TM",
    "actualTechnicalTimestamp": "2020-08-03T05:50:30.183Z",
    "actualBusinessTimeZone": "CET",
    "actualBusinessTimestamp": "2020-08-03T05:50:11Z",
    "locationAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:Location:ShippingPoint:0001",
    "myattachments": [
        {
             "fileName": "POD.png",
            "fileContentBase64": <Base64Code>
        },
        {
            "fileName": "PrivatePic.png",
            "privateAttachmentURL": "<URL to your private file>"
        }
    ]
}
```

Note

- Base64 is a binary-to-text encoding scheme that is commonly used to represent binary data (such as images or files) in a format that can be transmitted over text-based protocols, such as email or HTTP.

- The file specified within "`privateAttachmentURL`" is referenced by a URL that points to a file located within your internal network.
- In this example, "`myattachments`" is an event field with the type "Attachments" you defined in your model. You can have user-defined names. You can attach files by uploading them in base64 format, providing private attachment URLs, or using a combination of both methods.

Post an event with estimatedTimestamps

Here is the sample request body:

Sample Code

```
{
   "altKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVERY_IT:0080045552000040
   "quantity": 0.0,
   "messageSourceType": "TM",
   "actualTechnicalTimestamp": "2020-08-03T05:50:30.183Z",
   "actualBusinessTimeZone": "CET",
   "actualBusinessTimestamp": "2020-08-03T05:50:11Z",
   "locationAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:Location:ShippingPoint:0001",
   "estimatedTimestamps": [
       {
           "eventType": "com.gttsampleso01.gtt.app.sof.Delivery.LoadingStarted",
           "locationAltKey": "xri://sap.com/id:LBN#10010001016:QM7CLNT910:Location:ShippingPoin
           "estimatedTimestamp": "2020-08-03T06:50:11Z"
       }
   ]
}
```

Request Body Properties

Property	*Type*	*Mandatory*	*Description*
references	*Array of tracking Ids*	*no*	*The additional operation to tracking Ids of the tracked process.*
altKey	*String*	*yes*	*The alternative key.* *Assuming URN style keys we have as example first our EM-Style keys:* *xri://sap.com/id:LBN#10010001016:QM7CLNT910:OUTBOUND_DELIVI* *as string serialization with structural components:* • *xri://sap.com/id: This prefix is used to identify documents/proc system* • *LBN#10010001016: an id for solution owner or solution particip* • *QM7CLNT910: Business system name* • *OUTBOUND_DELIVERY: the document's type. This is TrackingId*

Property	*Type*	*Mandatory*	*Description*
			• *0080045552: The 'local' number range key*
validFrom	*Timestamp*	*no*	*The start date and time as local timestamp from which a tracking id is v particular tracked process. Supported timestamp format: yyyy-MM-ddT*
validTo	*Timestamp*	*no*	*The end date and time as local timestamp until when a tracking id is val tracked process. Supported timestamp format: yyyy-MM-ddTHH:mm:ss*
referenceType_code	*String*	*yes*	*Fixed: TRACKING*
action	*String*	*yes*	*Can be only set to ADD/DELETE* *ADD - creates/updates tracking id* *DELETE – deletes tracking id*
altKey	*String*	*yes*	*Same to references.altKey*
locationAltKey	*String*	*no*	*The alternative key of the location*
eventMatchKey	*String*	*no*	*Identifier for planned event*
eventReasonCode	*String*	*no*	*Pre-defined event reason code*
eventReasonText	*String*	*no*	*Event reason description*
messageSourceType	*String*	*no*	*Use the field to identify which source system the event comes from*
estimatedTimestamps	*Array of planned events with estimatedTimestamp*	*no*	*Information of estimated timestamp*
<Other custom parameter>		*no*	*Based on model definition*

Read Service

Description

- technical name: GTT_V2_OData_Query
- purpose: for access GTT tracked processes and events
- protocol: OData

General

Request URI

Request URI

- For HTTP basic authentication:

  https://<server address>/api/inbound/rest/v1/{model namespace}.{read service name}/

Example

https://*<server address>*/api/inbound/rest/v1/com.lbngttapps.gtt.app.gttft1.gttft1Service/

Response

Response Codes

The response code will be applied for all the following sample OData queries.

Code	HTTP Description	Track and Trace Semantic
200	OK	Query instance(s) have been retrieved
202	Accepted	If using OData batch mode, then this is the response
400	Bad Request	The URL is malformed
401	Unauthorized	There is no authentication passed with the request
403	Forbidden	The user does not have the right privileges to read the requested particular tracked process instance
404	Not Found	The requested particular tracked process instance was not found.
500	Internal Server Error	An unspecified error occurred during request processing

Read Metadata

Query for the metadata structure in the standard model "gttft1".

Request

Request URI

- For HTTP basic authentication:

  https://<server address>/api/outbound/odata/v1/{model namespace}.{read service name}/$metadata

  #### Example

  https://*<server address>*/api/outbound/odata/v1/com.lbngttapps.gtt.app.gttft1.gttft1Service/$metadata

HTTP Method

Get

Read Tracked Process List

Query for the Tracked Process List of the standard model "gttft1".

Request

Request URI

- For HTTP basic authentication:

  https://<server address>/api/outbound/odata/v1/{model namespace}.{read service name}/TrackedProcess

  Example

  https://*<server address>*/api/outbound/odata/v1/com.lbngttapps.gtt.app.gttft1.gttft1Service/TrackedProcess

HTTP Method

GET

Read Single Tracked Process Instance

Query for a specified Tracked Process of the standard model "gttft1".

Request

Request URI

- For HTTP basic authentication:

  https://<server address>/api/outbound/odata/v1/{model namespace}.{read service name}/{tracked process name}

  Example

  https://*<server address>*/api/outbound/odata/v1/com.lbngttapps.gtt.app.gttft1.gttft1Service/Shipment

HTTP Method

GET

Read Single Tracked Process Instance with Planned Events

Query for the Tracked Process with its planned events in the standard model "gttft1".

Request

Request URI

- For HTTP basic authentication:

  https://<server address>/api/outbound/odata/v1/{model namespace}.{read service name}/TrackedProcess('{uuid}')?$expand=plannedEvents

  Example

  https://*<server address>*/api/outbound/odata/v1/com.lbngttapps.gtt.app.gttft1.gttft1Service/TrackedProcess(guid'6d49b8f3-5599-5cbf-b9d0-b06bfb4e24a9')?$expand=plannedEvents

HTTP Method

GET

Read Planned Events of Single Tracked Process Instance

Query for the Tracked Process with its planned events in the standard model “gttfti1”.

Request

Request URI

- For HTTP basic authentication:

  https://<server address>/api/outbound/odata/v1/{model namespace}.{read service name}//TrackedProcess('{uuid}')/plannedEvents

  Example

  https://*<server address>*/api/outbound/odata/v1/com.lbngttapps.gtt.app.gttft1.gttft1Service/TrackedProcess(guid'6d49b8f3-5599-5cbf-b9d0-b06bfb4e24a9')/plannedEvents

HTTP Method

GET

Read Tracked Process Instances by Other Operators

The following table provides other operators that can be used to read tracked process instances: count, top, skip, orderby, filter, inlinecount, format.

Operator	Description	URI (Path, Service)
$count	Read the total count of Tracked Process instances	/api/outbound/odata/v1/<Model Namespace>.<Read Service Name>/TrackedProcess/$count
$skip	Read all Tracked Process instances except skip the first one	/api/outbound/odata/v1/<Model Namespace>.<Read Service Name>/TrackedProcess?$skip=1
$top	Read the first Tracked Process instance	/api/outbound/odata/v1/<Model Namespace>.<Read Service Name>/TrackedProcess?$top=1
$orderby	Read all Tracked Process instances ordered by ascending order	/api/outbound/odata/v1/<Model Namespace>.<Read Service Name>/TrackedProcess?$orderby=trackingId asc
$filter	Read the Tracked Process instances which trackingId equals “9900031705”	/api/outbound/odata/v1/<Model Namespace>.<Read Service Name>/TrackedProcess?$filter=trackingId eq '9900031705'
$inlinecount	Read the total count of Tracked Process instances and all Tracked Process instances	/api/outbound/odata/v1/<Model Namespace>.<Read Service Name>/TrackedProcess?$inlinecount=allpages
$format	Read all Tracked Process instances with json format	/api/outbound/odata/v1/<Model Namespace>.<Read Service Name>/TrackedProcess?$format=json

Current Supported OData Syntax and API Samples

The following tables lists OData syntax and API samples that are currently supported:

Query Option/ Operator		Description	Official Sample
$orderby			https://services.odata.org/OData/OData.svc/Products?$orderby=Rating
$top			https://services.odata.org/OData/OData.svc/Products?$top=5&$orderby=Name desc
$skip			https://services.odata.org/OData/OData.svc/Products?$skip=2&$top=2&$orderby=Rating
$count			https://services.odata.org/OData/OData.svc/Products/$count
$filter			
Logical Operators	Eq	Equal	/Suppliers?$filter=Address/City eq 'Redmond'
	Ne	Not equal	/Suppliers?$filter=Address/City ne 'London'
	Gt	Greater than	/Products?$filter=Price gt 20
	Ge	Greater than or equal	/Products?$filter=Price ge 10
	Lt	Less than	/Products?$filter=Price lt 20
	Le	Less than or equal	/Products?$filter=Price le 100
	And	Logical and	/Products?$filter=Price le 200 and Price gt 3.5
	Or	Logical or	/Products?$filter=Price le 3.5 or Price gt 200
	Not	Logical negation	/Products?$filter=not endswith(Description,'milk')
String Functions	Substringof(string p0, string p1)		https://services.odata.org/Northwind/Northwind.svc/Customers?$filter=substringof('Alfreds ', CompanyName) eq true
	endswith(string p0, string p1)		https://services.odata.org/Northwind/Northwind.svc/Customers?$filter=endswith(CompanyName , 'Futterkiste') eq true
	startswith(string p0, string p1)		https://services.odata.org/Northwind/Northwind.svc/Customers?$filter=startswith(CompanyName, 'Alfr') eq true
	tolower(string p0)		https://services.odata.org/Northwind/Northwind.svc/Customers?$filter=tolower(CompanyName) eq 'alfreds futterkiste'
	toupper(string p0)		https://services.odata.org/Northwind/Northwind.svc/Customers?$filter=toupper(CompanyName) eq 'ALFREDS FUTTERKISTE'
$expand			https://services.odata.org/OData/OData.svc/Categories?$expand=Products
			https://services.odata.org/OData/OData.svc/Products?$expand=Category,Suppliers
			https://services.odata.org/OData/OData.svc/Categories?$expand=Products/Suppliers